

Prime Factorisation

prime $\Rightarrow$ can be divided by 1 & itself.

e.g. 2, 3, 5, 7, 11, 13...

Express 30 as a product of its prime factors.

$$30 = 2 \times 3 \times 5$$

Express 220 as a product of its prime factors.

$$220 = 2^2 \times 5 \times 11$$

Highest Common Factor (HCF)

$\hookrightarrow$ largest factor that is common to all the numbers.

Find the highest common factor of 18 and 30.

Find the highest common factor of 80, 120 and 280.

lowest common multiple (LCM)

↳ smallest multiple that is common to all the numbers

pick the highest power.

↑

Find the lowest common multiple of 18 and 30.

Find the lowest common multiple of 5, 15 and 30.

Squares & Square Roots

- "perfect square" $\Rightarrow$ number whose square root is a whole number.
- square root $\Rightarrow \sqrt{a} / a^{\frac{1}{2}}$

Find the square root of 256 without using a calculator.

check-in:

Is the following a perfect square? :

a) 64 $\sqrt{64} = 8$ Yes.

b) 12 $\sqrt{12} = 2\sqrt{3}$ No.

c) 500 $\sqrt{500} = 10\sqrt{5}$ No.

d) 400 $\sqrt{400} = 20$ Yes.

Cubes and Cube Roots

$\downarrow$
 a^3

$\downarrow$
 $\sqrt[3]{a} / a^{\frac{1}{3}}$

Express 1188 as a product of its prime factors.

Answer _____ [1]

Using your answer to part (a), explain why 1188 is not a perfect cube.

$$1188 = 2^3 \times 3^3 \times 11$$

Since the power of the prime factor is not a multiple / cannot be divided by 3, it is not a perfect cube.

Ratio, Rate and Proportion

- $a : b$
- $\frac{a}{b}$

E.g. A book cost \$1.50. A pencil cost 90¢.

Express the ratio of their prices in the simplest form.

★ ensure that units are the same

Map scales

$\frac{\text{length}}{1} : r$
↑ ↑
1 cm on the map r cm on the actual piece of land

1 cm on the map

$\frac{\text{area}}{1} : r^2$
↑ ↑
1 cm² on the map r cm² on actual area of land.

In the map of a town, 10 km is represented by 5 cm.

- What is the actual distance if it is represented by a line of length 2 cm on the map?
- Express the map scale in the ratio 1 : n .
- Find the area of a plot of land that is represented by 10 cm².

$$\begin{array}{lcl}
 \text{a)} & 5\text{cm} & : 10\text{km} \\
 \div 5 \downarrow & 1\text{cm} & : 2\text{km} \\
 \times 2 \downarrow & 2\text{cm} & : 4\text{km}
 \end{array}
 \begin{array}{l}
 \downarrow \div 5 \\
 \downarrow \times 2
 \end{array}$$

$$1\text{ km} = 1000\text{ m} = 100\,000\text{ cm}$$

$$\therefore \text{Actual distance} = 4\text{ km}$$

$$\begin{array}{lcl}
 \text{b)} & 1\text{cm} & : 2\text{km} \\
 & 1 & : 200\,000
 \end{array}$$

$$\begin{array}{lcl}
 \text{c)} & (1\text{cm})^2 & : (2\text{km})^2 \\
 \times 10 \downarrow & 10\text{cm}^2 & : 40\text{km}^2 \\
 & 1\text{cm}^2 & : 4\text{km}^2
 \end{array}
 \begin{array}{l}
 \downarrow \times 10 \\
 \downarrow \times 10
 \end{array}$$

A length of 8 cm on a map represents an actual distance of 2 km. Find

- the actual distance represented by 25.6 cm on the map, giving your answer in km,
- the area on the map, in cm², which represents an actual area of 2.4 km²,
- the scale of the map in the form 1 : n .

$$\begin{array}{lcl}
 \text{a)} & 8\text{cm} & : 2\text{km} \\
 \div 8 \downarrow & 1\text{cm} & : 0.25\text{km} \\
 \times 25.6 \downarrow & 25.6\text{cm} & : 6.4\text{km}
 \end{array}
 \begin{array}{l}
 \downarrow \div 8 \\
 \downarrow \times 25.6
 \end{array}$$

$$\begin{array}{lcl}
 \text{c)} & 1\text{cm} & : 0.25\text{km} \\
 & 1 & : 25\,000
 \end{array}$$

$$\begin{array}{lcl}
 \text{b)} & 8\text{cm} & : 2\text{km} \\
 \div 2 \downarrow & 4\text{cm} & : 1\text{km} \\
 & (4\text{cm})^2 & : (1\text{km})^2 \\
 \times 2.4 \downarrow & 16\text{cm}^2 & : 1\text{km}^2 \\
 & 38.4\text{cm}^2 & : 2.4\text{km}^2
 \end{array}
 \begin{array}{l}
 \downarrow \div 2 \\
 \downarrow \times 2.4
 \end{array}$$

Q1. The scale of a map is 8 cm : 2 km .

(a) Write this scale in the form 1 : n .

Answer 1 : [1]

(b) The actual area of a lake is 90 000 m² .
Calculate the area, in square centimetres, of the lake on the map.

Answercm² [2]

Q2. A lake has an actual area of 2.56 km² . It is represented by an area of 4 cm² on a map.

(a) Find the scale of the map in the form 1 : n .

Answer 1 : [2]

(b) The distance between two towns on the map is 20 cm. Find the actual distance, in kilometres, between the two towns.

Answer km [1]

Written as the product of its prime factors, $126 = 2 \times 3^2 \times 7$.

- (a) (i) Express 600 as the product of its prime factors.

Answer: [1]

- (ii) Hence, find the LCM of 126 and 600.

Answer: [1]

- (b) Find the smallest positive integer k such that $\frac{126}{k}$ is a perfect square.

Answer: [1]